Basics & Syntax
1. What is Python? Mention two of its key features
python is a high level interpreted language.
features-1. Simple and Easy to Learn
2. Interpreted Language

2. How do you write a single-line comment in Python?
with #

3. How do you write a multi-line comment in Python?
""" """ using to write multi line comment.

4. What is the difference between print() and return?
print() — Displays Output to the Console
return — Sends Output Back to the Caller

5. How do you get user input in Python?
You can get input from the user using the input() function

6. How do you check the version of Python installed?
Method 1: Using Python Code
Method 2: Using the Command Line (Terminal)

7. Is Python case-sensitive? Give an example.
Yes, Python is case-sensitive.
This means that identifiers (like variable names, function names, class names, etc.) with different cases are treated as different entities.
Example:

```
name = "Alice"
Name = "Bob"

print(name)  # Outputs: Alice
print(Name)  # Outputs: Bob
```

8. How to Run a Python Script from the Command Line

To run a Python script using the command line (also called terminal or shell), follow these steps:
Step 1: Save Your Python Script
step 2: Open the Command Line
Step 3: Navigate to the Script's Directory

9. What are keywords in Python? How can you list them?
keywords are reserved words in python.
if, else, elif

for, while, break, continue

def, return

class, try, except, finally
10. How do you declare a variable in Python?
In Python, declaring a variable is very simple — you just assign a value to a name using the = operator.

11. What’s the difference between = and ==?
= is a assignment operator to using the assign valued
== is a comparison operator to compare the value

12. How do you swap two variables without using a third variable?
Python makes swapping values simple and elegant — you can do it in one line using tuple unpacking (no third variable needed).
a, b = b, a
13. How do you write a one-line if statement?
if x>10:print("x is greater than 10")

14. What’s the difference between None and 0 in Python?
None- special constant meaning "no"
0-integer valued zero (numeric)
15. What is indentation in Python and why is it important?
Indentation in Python means using spaces or tabs at the beginning of a line to define blocks of code
It is important because Python uses indentation to determine the structure and grouping of statements

2. Data Types & Operations (16–35)

16. What are Python’s built-in data types?

Common built-in data types include:

Numeric: int, float, complex

Sequence: list, tuple, range
Text: str
Mapping: dict
Set types: set, frozenset
Boolean: bool
NoneType: None

17. How do you check the type of a variable?

Use the built-in type() function:
```
x = 5
print(type(x))
```
18. What’s the difference between a list and a tuple?
list=Mutable, dynamic data, slightly slower
tuple=Immutable, fixed size data, faster
19. How do you create a dictionary in Python?
```
my_dict = {
    "name": "Alice",
    "age": 30,
}
```

20. What’s the difference between append() and extend() for lists?
append(item) — adds a single item to the end of the list.
extend(iterable) — adds each element of an iterable to the end.

21. How do you remove an item from a list?
By value: list.remove(value) (removes first occurrence)
By index: del list[index] or list.pop(index)

22. How Do You Reverse a List in Python?
You can reverse a list in Python in two common ways:
```
my_list.reverse()
#in-place
reversed_list =
my_list[: : -1] #COPY
#returns an iterator
```
23. How do you sort a list in ascending order?
Use the .sort() method or sorted() function
24. What is the difference between shallow copy and deep copy?
Shallow copy: Copies the outer object only.Nested objects are still shared.
Deep copy: Recursively copies all nested objects — completely independent
25. How do you convert a string to lowercase?
Use .lower() method:
26. How do you check if a string starts with a particular word?
Use .startswith() method
27. What’s the difference between is and ==?
is-Check objet identity(same memory location)
==- check value equality
28. How do you merge two dictionaries in Python 3.9+?
dict3=dict1|dict2
29. How do you find the length of a dictionary?
(len(my_dict))
30. How do you create a set?
my_set = set([1, 2, 3])

31. What’s the difference between set() and {} in Python?
set() → creates an empty set
{} → creates an empty dictionary

32. How do you find the union of two sets?
set1.union(set2)

33. How do you find the intersection of two sets?
set1.intersection(set2)

34. What’s the difference between remove() and discard() in sets?
Both are used to delete elements from a set, but:
remove(x) → Removes element x.
If x is not present, it raises a KeyError
discard(x) → Removes element x.
If x is not present, it does nothing (no error).

35. How do you convert a list into a tuple?
You can use the tuple() constructor.

3. Control Flow
36. How does the if-elif-else structure work in Python?
if x > 0:

```
if x > 0:
  print("x is positive")
elif x < 0:
  print("x is negative")
else:
  print("x is zero")
37. What is the difference between for and while loops?
for → Iterates over a sequence.
while → Runs until a condition becomes false.
38. How do you loop through a dictionary’s keys and values?
for key, value in my_dict.items():
    print(key, value)
39. How do you break out of a loop?
break
40. How do you skip the current iteration in a loop?
continue
41. What is an infinite loop? Example:
Loop that never ends:
while True:
    print("Hello")
42. How do you use the range() function?
range(start, stop, step)

43. Difference between range(5) and range(1,5)?
range(5) → 0 to 4.
range(1,5) → 1 to 4.

44. How do you iterate over both index and value in a list?
for index, value in enumerate(my_list):
    print(index, value)
45. How does the else clause work with loops?

Executes if the loop completes normally (no break).

46. How do you use a nested loop?
for i in range(3):
    for j in range(2):
        print(i, j)
47. How do you loop through multiple lists simultaneously?
for a, b in zip(list1, list2):
    print(a, b)
48. How do you reverse iterate over a list?
for item in reversed(my_list):
    print(item)
49. What is a list comprehension? Example:
Compact syntax for creating lists.
squares = [x*x for x in range(5)]
50. How do you use a conditional inside a list comprehension?
even_numbers = [x for x in range(10) if x % 2 == 0]

4.FUNCTIONS
51. How do you define a function in Python?
In Python, you define a function using the def keyword
def function_name(parameters)

52.Difference between positional and keyword arguments
Positional Arguments → Values are passed in the same order as parameters are defined.
Keyword Arguments → Values are passed by explicitly naming the parameter, order does not matter

53. Default parameter values in functions
A parameter can have a default value, which is used if no argument is provided for that parameter.

54. Passing a variable number of arguments
Python provides two special symbols:
*args → allows passing a variable number of positional arguments (tuple).
**kwargs → allows passing a variable number of keyword arguments (dictionary).

55. How do you return multiple values from a function?
In Python, a function can return multiple values as a tuple.
def calc():
    return 1, 2
a, b = calc()

56. What is a lambda function? Give an example.
A lambda function is an anonymous (nameless), single-line function defined using the lambda keyword.

57. What’s the difference between local and global variables?
Local variable → defined inside a function, accessible only within that function.
```

Global variable → defined outside all functions, accessible throughout the program.

58. How do you modify a global variable inside a function?
Use the global keyword.

59. What is recursion in Python? Give an example.
Recursion = a function calling itself to solve a smaller version of the problem.
Example: Factorial using recursion

60. What is a docstring in Python functions?
A docstring is a string literal written just below the function definition to describe its purpose.

61. How do you use type hints in functions?
Type hints specify expected types of parameters and return values.

62. What are function annotations?
Function annotations = metadata about function parameters and return values (same as type hints, but can be any expression).

63. What is the purpose of the pass statement?
pass is a placeholder statement that does nothing.

5.OOPS
66. How do you define a class in Python?
A class is defined using the class keyword

67. How do you create an object from a class?
An object is created by instantiating the class (calling it like a function).

68. What is __init__ in Python?
__init__ is a constructor method in Python.
It runs automatically when an object is created.

69. Difference between instance variables and class variables
INSTANCE VARIABLE-inside methods using self
CLASS VARIABLE-directly inside class but not outside method

70. What is inheritance in Python?
Inheritance = process where one class (child) derives properties & methods from another class (parent).
71. How do you call a parent class constructor?
You use the super() function or explicitly call the parent class name.

73. Difference between method overloading and overriding
Overloading → Same method name, different parameters (not natively supported in Python).
Overriding → Child class changes parent method.

74. What is multiple inheritance?
When a class inherits from more than one parent class.

75. What’s the role of super() in Python?
super() is used to call methods of the parent class.
Commonly used in constructors to initialize parent attributes.

76. What are magic methods in Python? Give two examples.
Magic methods = special methods in Python that begin and end with __.
Examples:
__init__ → constructor (initialization).
__add__ → defines behavior of + operator.

77. What does __str__ do in a class?
__str__ defines the string representation of an object when printed.

78. What is polymorphism in Python?
Polymorphism = same function/method/operator behaves differently depending on the object.

79. What is encapsulation in Python?
Encapsulation = binding data (attributes) and methods into a single unit (class), and restricting direct access to some attributes

80. What are @staticmethod?
A staticmethod is a method inside a class that doesn’t use self or cls.

81.How do you make an attribute private in Python?
Use double underscore (__) before the attribute name.

7.Modules, Packages & File Handling
89. How do you import a module in Python?
In Python, you use the import statement to bring in a module so you can use its functions, classes, and variables.

```
90. Difference between import module and from module import?
import math → Must use math.sqrt()
from math import sqrt → Use sqrt() directly.

91. How do you find the location of an imported module?
import math
print(math.__file__)

 92. How do you create your own module in Python?
Create a .py file with functions.
Import it in another script with import
```

Could not connect to the reCAPTCHA service. Please check your internet connection and reload to get a reCAPTCHA challenge.